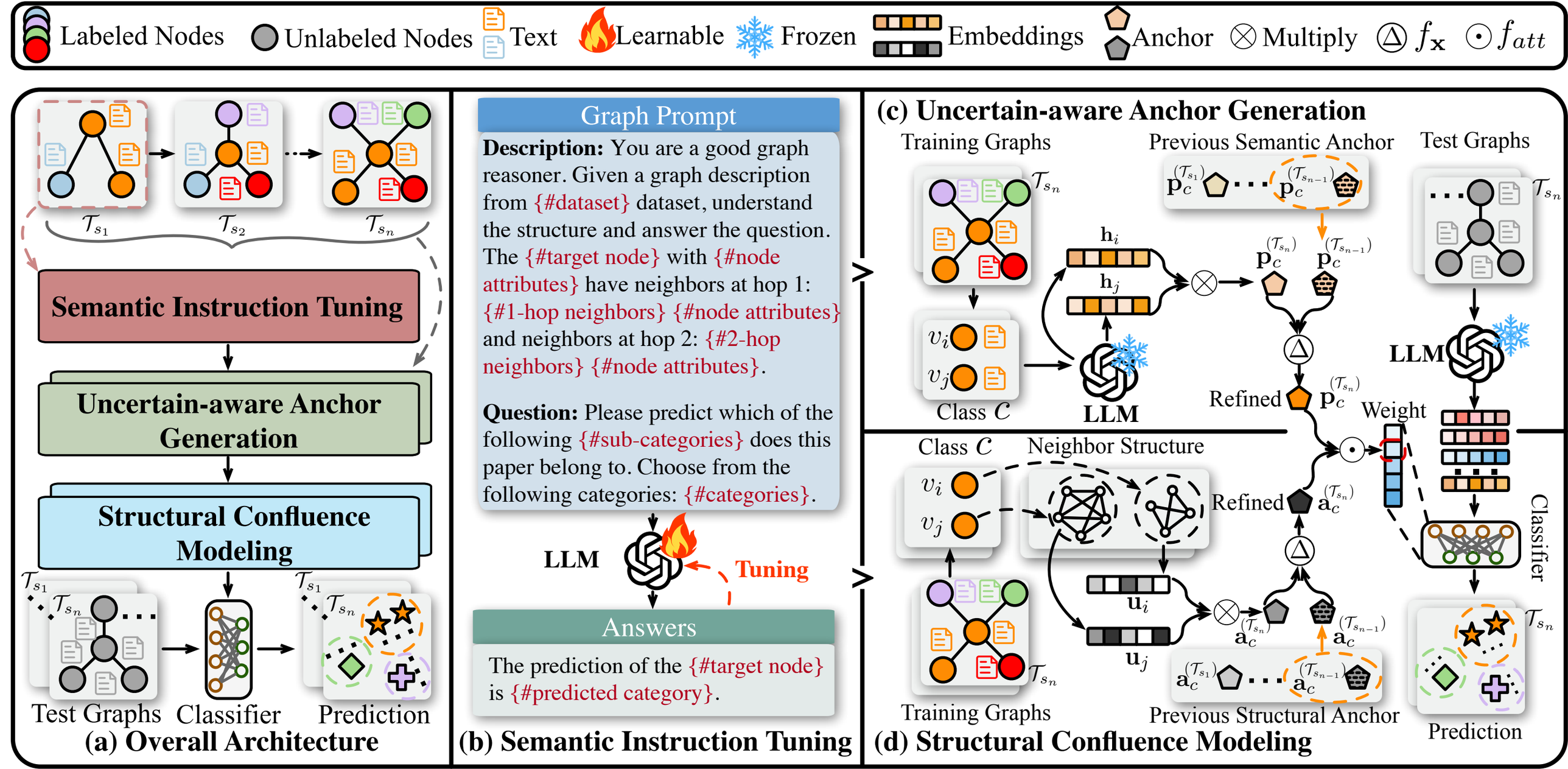

Labeled Nodes
Unlabeled Nodes
Text
Learnable
Frozen
Embeddings
Anchor
Multiply
$f_{\mathbf{x}}$
f_{att}
$\mathcal{T}_{s_1}$
$\mathcal{T}_{s_2}$
$\mathcal{T}_{s_n}$
Semantic Instruction Tuning
Uncertain-aware Anchor Generation
Structural Confluence Modeling
Test Graphs
Classifier
Prediction
(a) Overall Architecture
Graph Prompt
Description: You are a good graph reasoner. Given a graph description from {#dataset} dataset, understand the structure and answer the question. The {#target node} with {#node attributes} have neighbors at hop 1: {#1-hop neighbors} {#node attributes} and neighbors at hop 2: {#2-hop neighbors} {#node attributes}.
Question: Please predict which of the following {#sub-categories} does this paper belong to. Choose from the following categories: {#categories}.
LLM
Tuning
Answers
The prediction of the {#target node} is {#predicted category}.
(b) Semantic Instruction Tuning
(c) Uncertain-aware Anchor Generation
Training Graphs
Previous Semantic Anchor
Test Graphs
Class C
LLM
Refined
Weight
Classifier
Neighbor Structure
Training Graphs
Previous Structural Anchor
Prediction
(d) Structural Confluence Modeling